

# APPLICATIONS OF INTEGRATION

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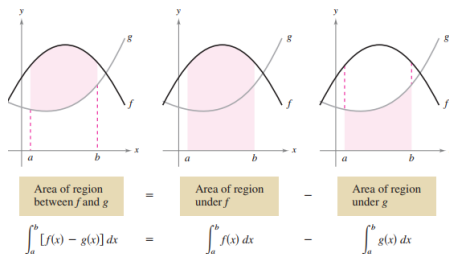
# Lecture Outline

- 1 Area of a Region Between Two Curves
- 2 Volume of A solid
  - Disk Method
  - Washer Method
  - Volume of A solid With Known Cross Sections
- 3 Arc Length and Surfaces of Revolution
- 4 Work and Fluid Forces



## Area of a Region Between Two Curves

We can extend the application of definite integrals from the area of a region under a curve to the area of a region between two curves. Consider two functions that are continuous on the interval  $[a, b]$



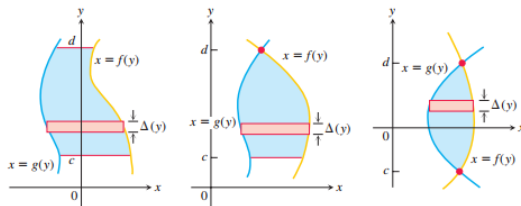
If  $f$  and  $g$  are continuous on  $[a, b]$  and  $g(x) \leq f(x)$  for all  $x$  in  $[a, b]$  then the area of the region bounded by the graphs of  $f$  and  $g$  and the vertical lines  $x = a$  and  $x = b$  is

$$A = \int_a^b [f(x) - g(x)] dx = \int_{a=x_1}^{b=x_2} [\text{top curve} - \text{bottom curve}] dx$$



# Area of a Region Between Two Curves

If a region's bounding curves are described by functions of  $y$ , the approximating rectangles are horizontal instead of vertical and the basic formula has  $y$  in place of  $x$ .



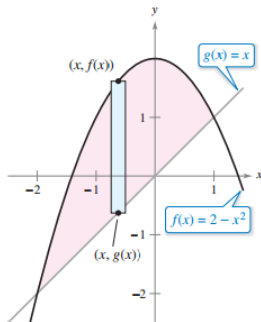
$$A = \int_c^d [f(y) - g(y)] dy = \int_{c=y_1}^{d=y_2} [\text{right curve} - \text{left curve}] dy$$



# Area of a Region Between Two Curves

## Example

Find the area of the region bounded by the graphs of  $f(x) = 2 - x^2$  and  $g(x) = x$



The plot have two intersections points; that is the limits for integration. First equate  $f(x)$  to  $g(x)$  to find these points.

$$2 - x^2 = x \implies x^2 + x - 2 = 0 \implies x = -2, \text{ or } 1 \quad (1)$$

Then the area is given as

$$A = \int_a^b [f(x) - g(x)] dx \quad (2)$$

$$= \int_{-2}^1 [(2 - x^2) - x] dx \quad (3)$$

$$= -\frac{x^3}{3} - \frac{x^2}{2} + 2x \Big|_{-2}^1 = \frac{9}{2} \quad (4)$$



# Area of a Region Between Two Curves

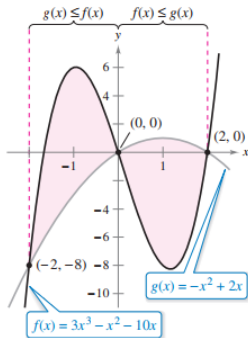
## Example

Find the area of the region between the graphs of  $f(x) = 3x^3 - x^2 - 10x$  and  $g(x) = -x^2 + 2x$

Again equate  $f(x)$  to  $g(x)$  to find intersection points.

$$\begin{aligned} 3x^3 - x^2 - 10x &= -x^2 + 2x \implies 3x^3 - 12x = 0 \\ \implies 3x(x-2)(x+2) &= 0 \implies x = -2, 0, 2 \end{aligned}$$

Note  $g(x) \leq f(x)$  when  $x \in [-2, 0]$  and  $f(x) \leq g(x)$  when  $x \in [0, 2]$  Then the area is given as



$$A = \int_{-2}^0 [f(x) - g(x)] dx + \int_0^2 [g(x) - f(x)] dx \quad (5)$$

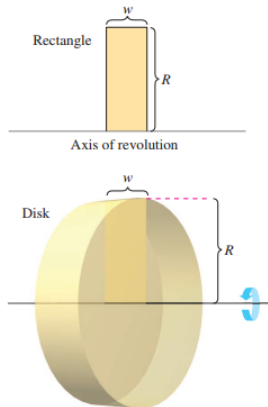
$$= \int_{-2}^0 (3x^3 - 12x) dx + \int_0^2 (-3x^3 + 12x) dx \quad (6)$$

$$= \left. \frac{3x^4}{4} - 6x^2 \right|_{-2}^0 + \left. \frac{-3x^4}{4} + 6x^2 \right|_0^2 = 24$$



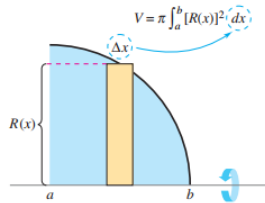
## Volume of A solid: Disk Method

When a region in the plane is revolved about a line, the resulting solid is a solid of revolution, and the line is called the axis of revolution.

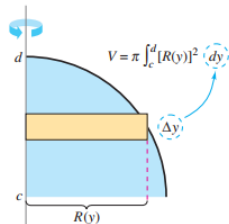


To find the volume of a solid of revolution with the disk method, use one of the formulas below:

$$V = \pi \int_a^b [R(x)]^2 dx \text{ or } V = \pi \int_c^d [R(y)]^2 dy$$



Horizontal axis of revolution

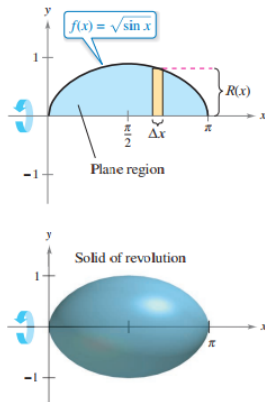


Vertical axis of revolution



## Example

Find the volume of the solid formed by revolving the region bounded by the graph of  $f(x) = \sqrt{\sin x}$  and the x-axis  $x \in [0, \pi]$  about the x-axis.



$$V = \pi \int_a^b [R(x)]^2 dx \quad (8)$$

$$= \pi \int_0^\pi [\sqrt{\sin x}]^2 dx \quad (9)$$

$$= \pi \int_0^\pi \sin x \, dx \quad (10)$$

$$= \pi (-\cos x) \Big|_0^\pi \quad (11)$$

$$= \pi(1 + 1) \quad (12)$$

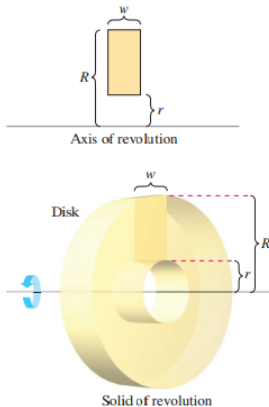
$$= 2\pi \quad (13)$$





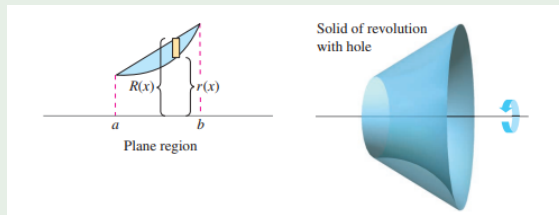
## Volume of A solid: Washer Method

The disk method can be extended to cover solids of revolution with holes by replacing the representative disk with a representative washer. The washer is formed by revolving a rectangle about an axis with inner  $r(x)$  and outer radii  $R(x)$ .



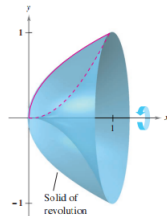
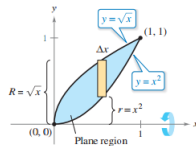
The volume of a solid of revolution using the washer method is:

$$V = \pi \int_a^b \left[ [R(x)]^2 - [r(x)]^2 \right] dx$$



## Example

Find the volume of the solid formed by revolving the region bounded by the graphs of  $y = \sqrt{x}$  and  $y = x^2$  about the x-axis.



$$V = \pi \int_a^b \left[ [R(x)]^2 - [r(x)]^2 \right] dx \quad (14)$$

$$= \pi \int_0^1 \left[ (\sqrt{x})^2 - (x^2)^2 \right] dx \quad (15)$$

$$= \pi \int_0^1 (x - x^4) dx \quad (16)$$

$$= \pi \left[ \frac{x^2}{2} - \frac{x^5}{5} \right] \bigg|_0^1 \quad (17)$$

$$= \frac{3\pi}{10} \quad (18)$$

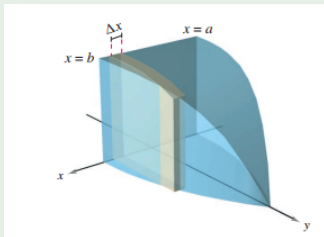


## Volume of A solid With Known Cross Sections

With the disk method, you can find the volume of a solid having a circular cross section. This method can be generalized to solids of any shape, as long as you know a formula for the area of an arbitrary cross section.

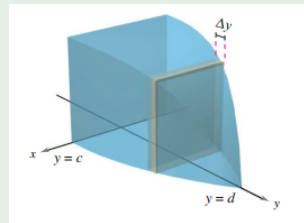
For cross sections of area  $A(x)$  taken perpendicular to the  $x$ -axis

$$V = \int_a^b A(x) dx$$



For cross sections of area  $A(y)$  taken perpendicular to the  $y$ -axis

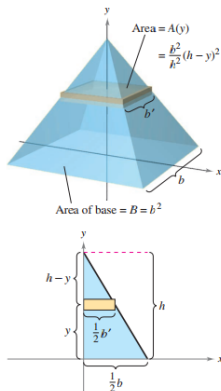
$$V = \int_c^d A(y) dy$$



## Example

Prove that the volume of a pyramid with a square base is  $V = \frac{1}{3}hB$  where  $h$  is the height of the pyramid and  $B$  is the area of the base.

As shown, you can intersect the pyramid with a plane parallel to the base at height  $y$  to form a square cross section whose sides are of length  $b'$ .



Using similar triangles  $\frac{b'}{b} = \frac{h-y}{h} \implies b' = \frac{b}{h}(h-y)$  so the Area of the square cross section is  $A(y) = (b')^2 = \frac{b^2}{h^2}(h-y)^2$

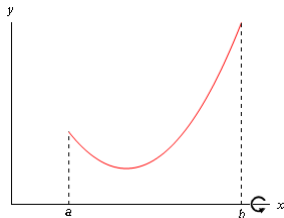
$$V = \int_0^h A(y) dy = \int_0^h \frac{b^2}{h^2} (h-y)^2 dy \quad (19)$$

$$= \frac{b^2}{h^2} \int_0^h (h-y)^2 dy = -\frac{b^2}{h^2} \left[ \frac{(h-y)^3}{3} \right] \Big|_0^h \quad (20)$$

$$= \frac{b^2}{h^2} \left[ \frac{h^3}{3} \right] = \frac{1}{3}hB, \quad \text{where } B = b^2 \quad (21)$$

# Arc Length

To determine the length of the continuous function  $y = f(x)$  on the interval  $[a, b]$ . We'll also need to assume that the derivative is continuous on  $[a, b]$ .



The length of this curve is given as

$$L = \int ds \quad (22)$$

where,

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{if } y = f(x), a \leq x \leq b \quad (23)$$

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \quad \text{if } x = h(y), c \leq y \leq d \quad (24)$$

Using the first  $ds$  will require  $x$  limits of integration and using the second  $ds$  will require  $y$  limits of integration. The denominator of the derivative is always the derivative of the integral.



# Arc Length

## Example

Determine the length of  $y = \ln(\sec x)$ ; where  $0 \leq x \leq \frac{\pi}{4}$

$$y = f(x) = \ln(\sec x) \Rightarrow \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x \Rightarrow \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$

Therefore

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x$$

The arc length is then,

$$\begin{aligned} L &= \int_0^{\frac{\pi}{4}} \sec x \, dx \\ &= \ln|\sec x + \tan x| \Big|_0^{\frac{\pi}{4}} \\ &= \ln(\sqrt{2} + 1) \end{aligned}$$



# Arc Length

## Example

Determine the length of  $x = \frac{1}{2}y^2$ ; for  $0 \leq x \leq \frac{1}{2}$ . Assume that  $y$  is positive.

$$\frac{dx}{dy} = y \quad \Rightarrow \quad \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + y^2}$$

Before writing down the length notice that we were given  $x$  limits, and we will need  $y$  limits for this  $ds$ . With the assumption that  $y$  is positive these are easy enough to get. All we need to do is plug  $x$  into our equation and solve for  $y$ . Doing this gives,

$$0 \leq y \leq 1$$

The integral for the arc length is then,

$$L = \int_0^1 \sqrt{1 + y^2} dy$$



This integral will require the following trigonometry substitution.

$$y = \tan \theta \qquad dy = \sec^2 \theta \, d\theta$$

$$y = 0 \qquad \Rightarrow \qquad 0 = \tan \theta \qquad \Rightarrow \qquad \theta = 0$$

$$y = 1 \qquad \Rightarrow \qquad 1 = \tan \theta \qquad \Rightarrow \qquad \theta = \frac{\pi}{4}$$

$$\sqrt{1 + y^2} = \sqrt{1 + \tan^2 \theta} = \sqrt{\sec^2 \theta} = \sec \theta$$

The length is then,

$$\begin{aligned} L &= \int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta \\ &= \frac{1}{2} (\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|) \Big|_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left( \sqrt{2} + \ln(1 + \sqrt{2}) \right) \end{aligned}$$





# Surfaces of Revolution

Below is a sketch of a function  $y = f(x)$  in the interval  $[a, b]$  and the solid of revolution we get by rotating the function about the x-axis.

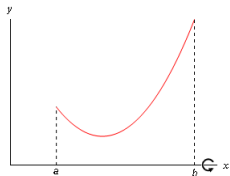


Figure A

The surface area of this solid (Figure B) is obtained using the formulas:

$$S = \int 2\pi y \, ds \quad \text{rotation about } x\text{-axis} \quad (25)$$

$$S = \int 2\pi x \, ds \quad \text{rotation about } y\text{-axis} \quad (26)$$

where,

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx \quad \text{if } y = f(x), \, a \leq x \leq b \quad (27)$$

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy \quad \text{if } x = h(y), \, c \leq y \leq d \quad (28)$$

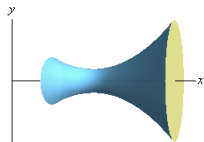


Figure B



## Example

Determine the surface area of the solid obtained by rotating  $y = \sqrt{9 - x^2}$ ,  $-2 \leq x \leq 2$  about the x-axis.

The formula that we will be using here is,  $S = \int 2\pi y ds$

$$\frac{dy}{dx} = \frac{1}{2}(9 - x^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{(9 - x^2)^{\frac{1}{2}}}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{x^2}{9 - x^2}} = \sqrt{\frac{9}{9 - x^2}} = \frac{3}{\sqrt{9 - x^2}}$$

$$S = \int_{-2}^2 2\pi y \frac{3}{\sqrt{9 - x^2}} dx$$

Before evaluating the integral we'll need to substitute in for y as well.

The surface area is then,

$$\begin{aligned} S &= \int_{-2}^2 2\pi \sqrt{9 - x^2} \frac{3}{\sqrt{9 - x^2}} dx \\ &= \int_{-2}^2 6\pi dx = 24\pi \end{aligned}$$



# Work

The work done by a variable force  $F(x)$  in moving an object along the  $x$ -axis from  $x = a$  to  $x = b$  is

$$W = \int_a^b F(x) dx$$

## Hooke's Law

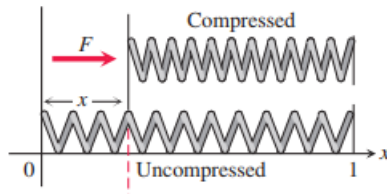
Hooke's law says that the force required to hold a stretched or compressed spring  $x$  units from its natural (unstressed) length is proportional to  $x$ . In symbols  $F = kx$



# Work

## Example

Find the work required to compress a spring from its natural length of  $1\text{ ft}$  to a length of  $0.75\text{ ft}$  if the force constant is  $k = 16$ .



The work done by  $F$  over this interval is

$$W = \int_0^{0.25} 16x dx = 8x^2 \Big|_0^{0.25} = 0.5$$



## Exercise

1 Find the areas of the regions enclosed by the following curves

1  $y = x^2 - 2$  and  $y = 2$

2  $y = 2x - x^2$  and  $y = -3$

3  $y = x^4$  and  $y = 8x$

4  $y = 7 - 2x^2$  and  $y = x^2 + 4$

5  $x = 2y^2$ ,  $x = 0$ , and  $y = 3$

6  $y^2 - 4x = 4$  and  $4x - y = 16$

7  $x + y^2 = 0$  and  $x + 3y^2 = 2$

2 Find the volume of the solid generated by revolving the region bounded by the graphs of the equations about the x-axis

1  $y = \frac{1}{x}$ ,  $y = 0$ ,  $x = 1$ ,  $x = 3$

2  $y = e^{-x}$ ,  $y = 0$ ,  $x = 0$ ,  $y = 1$

3  $y = x^2 + 1$ ,  $y = -x^2 + 2x + 5$ ,  $x = 0$ ,  $x = 3$

3 Find the volume of the solid generated by revolving the region bounded by the graphs of the equations about the y-axis

1  $y = 3(2 - x)$ ,  $y = 0$ ,  $x = 0$

2  $y = 9 - x^2$ ,  $y = 0$ ,  $x = 2$ ,  $x = 3$

4 Use the disk method to verify that the volume of a sphere is  $\frac{4}{3}\pi r^3$ , where  $r$  is the radius.

## Exercise

- 1 Determine the length of

$$y = \sqrt{x+2}, \quad 1 \leq x \leq 7$$

using,

$$1 \quad ds = \sqrt{1 + \left[ \frac{dy}{dx} \right]^2} dx$$

$$2 \quad ds = \sqrt{1 + \left[ \frac{dx}{dy} \right]^2} dy$$

- 2 Determine the length of  $y = 7(6+x)^{\frac{3}{2}}$ ,  $189 \leq y \leq 875$ .

- 3 Determine the length of  $x = 4(3+y)^2$ ,  $1 \leq y \leq 4$ .

- 4 Find the surface area of the object obtained by rotating

$$y = 4 + 3x^2, \quad 1 \leq x \leq 2$$

about the y-axis.

- 5 Find the surface area of the object obtained by rotating

$$y = \sin(2x), \quad 0 \leq x \leq \frac{\pi}{8}$$

## Exercise

- ① A force of 5 pounds compresses a 15-inch spring a total of 3 inches. How much work is done in compressing the spring 7 inches?
- ② A force of 250 newtons stretches a spring 30 centimeters. How much work is done in stretching the spring from 20 centimeters to 50 centimeters?
- ③ A force of 20 pounds stretches a spring 9 inches in an exercise machine. Find the work done in stretching the spring 1 foot from its natural position.
- ④ An overhead garage door has two springs, one on each side of the door. A force of 15 pounds is required to stretch each spring 1 foot. Because of the pulley system, the springs stretch only one-half the distance the door travels. The door moves a total of 8 feet, and the springs are at their natural length when the door is open. Find the work done by the pair of springs.



END OF LECTURE  
THANK YOU

